

GREEN TECHNOLOGIES

THE METHODS THAT CAN CLEAN UP THE PLANET

Aditya Madanapalle | aditya@digit.in

There are three primary areas of concern where it is crucial for technologies and methods to prevent further deterioration of the global ecosystem. These are the removal of greenhouse gases from the atmosphere, the removal of plastic waste, particularly from the oceans, and preventing the ice caps, as well as the permafrost from melting.

CARBON SEQUESTRATION

While there are global efforts to reduce the amount of carbon dioxide emitted into the atmosphere, a number of emerging technologies are focusing on artificial ways to remove the carbon already dumped into the atmosphere. The first carbon capture plant was proposed in 1938, and was first implemented in 1972 at the Ridge oilfield in Texas. The idea is pretty simple, the carbon dioxide released from industrial sources is captured at the point of emission, and funneled to deep deposits within the Earth, such as depleted oil reservoirs or unmineable coal seams. There are over forty large scale carbon capture and storage facilities around the world. However, all these facilities are together just capturing one percent of the carbon that is necessary to meet the conditions of the Paris agreement by 2040. The captured carbon dioxide can actually make it easier to extract oil from the ground. Instead of depleted reservoirs,

the carbon is injected into existing oil reserves, making it easier for energy companies to extract the oil. The carbon dioxide then remains within the reservoirs, replacing the oil.

A physicist from Harvard, David W. Keith has started a company funded by Bill Gates called Carbon Engineering. The company has developed technologies to capture carbon dioxide directly from the air. Essentially, fans are used to run the air through chemical processes, where the carbon is passed over a solvent. The technology developed by



The clean burning fuel on the right by Carbon Engineering

Carbon Engineering just needs water and energy, with the output being a pure stream of compressed carbon dioxide. The company has proved their capturing technology by running a plant for a year, extracting a tonne of pure carbon dioxide for every day of the year. The collected carbon was released back to the atmosphere, but Carbon En-

gineering has another innovation up its sleeve that can put the carbon dioxide to good use. The idea is to use the captured carbon dioxide to make a clean burning fuel, that can be used to power everything from cars to airplanes. This patented and trademarked technology is called Air to Fuels, and is a proposed replacement for crude oil. The fuel is compatible with existing vehicles, and can be used without any further modifications. Additionally, the fuel can be blended with traditional fuels to any extent, to allow for progressive reduction in emissions. While electrifying cars is viable, aircraft and ships need the high energy density of liquid fuel, and Carbon Engineering hopes to serve that need with its Air to Fuels technology.

Researchers from the Earth Institute at Columbia University, have developed what they call "synthetic trees" to remove carbon dioxide from the atmosphere at faster rates than natural trees. These synthetic trees can line highways or form forests on their own, and be used to extract carbon dioxide from the atmosphere at massive scales. The trees consume very little energy during the carbon capture process, and use proprietary technology to passively absorb carbon dioxide from the air – no fans. The expected life span of each tree is about 15 years, and captured gas can be sold as a nutrient for farms, used to make carbonated drinks, or dry ice.